

Project Proposal and Preliminary Design

Project Title: Real-Time FPGA-Based Gesture Recognition System

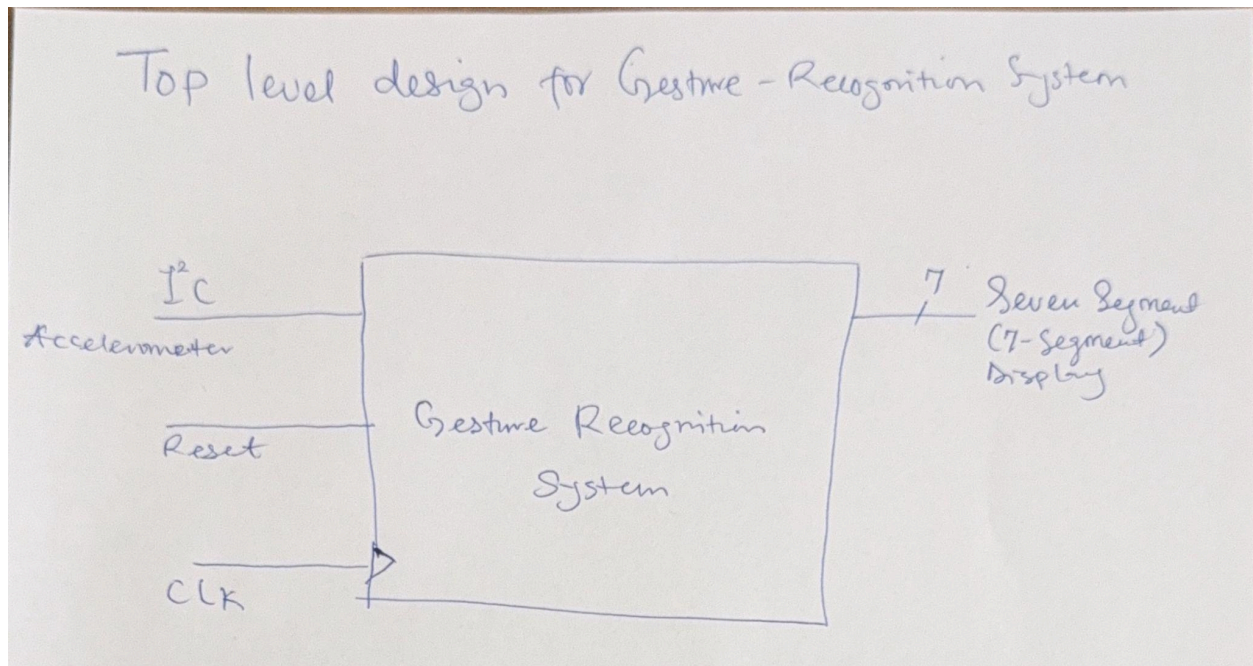
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Innocent Onwukanjo

1. General Description

This project is the design and implementation of a real-time FPGA-based gesture recognition system that receives spatial data through the onboard I2C accelerometer and translates physical tilts of the board into visual outputs on the 7-segment display. For the advanced feature set, we are using the GPIO pins.

2. Top-Level Diagram (Inputs and Outputs)

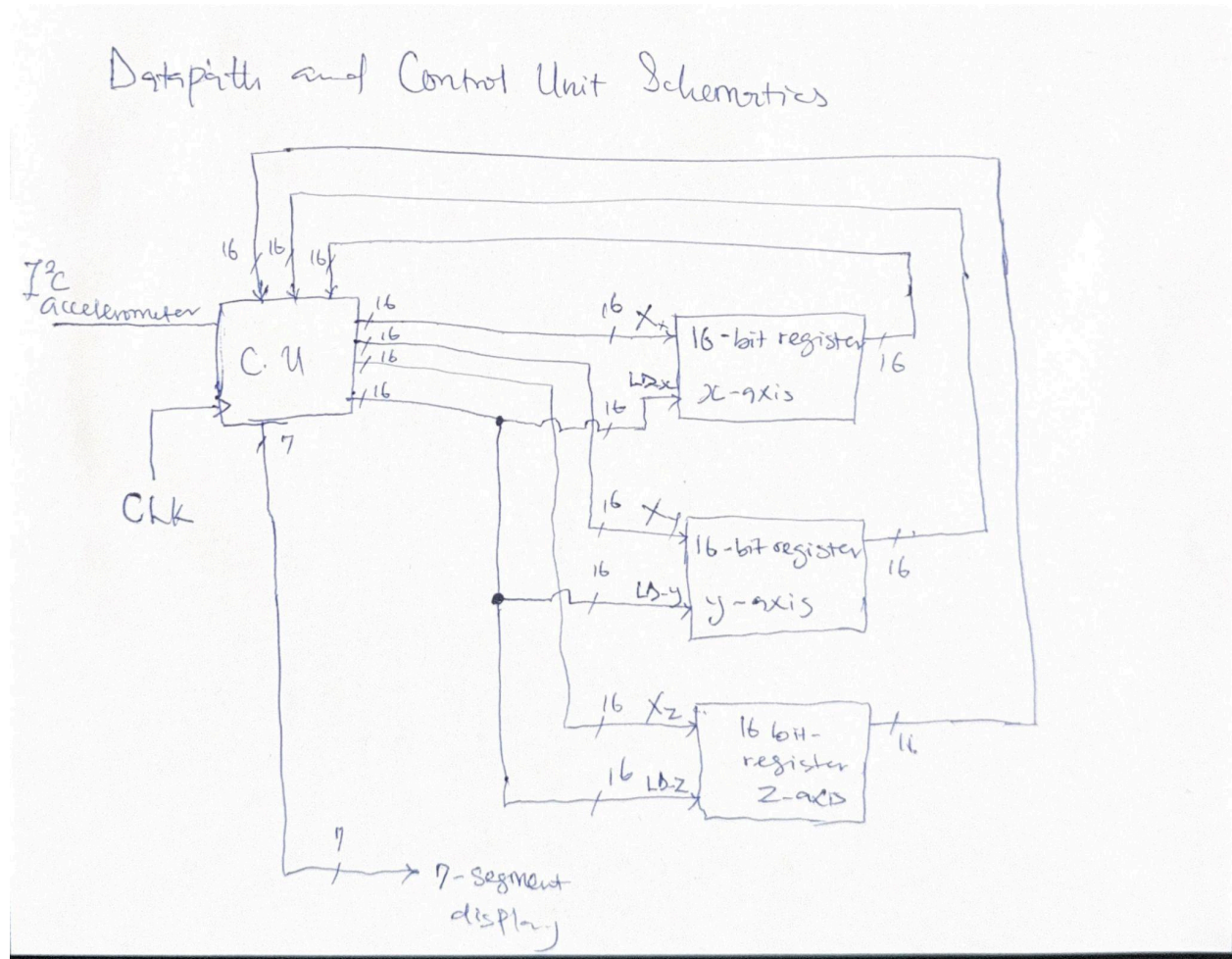


The System Black Box.

The input to the system is the I2C Accelerometer, Synchronous Reset, and the Clock.

The output to the system is the 7-segment display

3. Preliminary Datapath and Control Signals



The Data Path and Control Unit

The datapath consists of three parallel 16-bit registers dedicated to storing the incoming X, Y, and Z axis data.

The Control Unit operates as a Finite State Machine that manages the system timing and contains embedded magnitude comparators to evaluate if the incoming tilt values cross the +/- 200 thresholds

To synchronize these operations, the Control Unit sends load signals (LD) to latch data into the datapath registers, which then route the stored 16-bit values back up to the Control Unit for mathematical computation.

4. Major Test Cases and Test Plan

We will implement a self-checking testbench with automated verification using VHDL assertions to validate the data.

Major test cases will include simulating boundary data values exactly around the ± 200 thresholds (such as 199 and 201), as well as a neutral zero state, to verify the FSM comparators accurately decode gestures without false positive triggers.

5. Demonstration Capabilities

We will demonstrate the system by physically tilting the DE10-Standard board to trigger specific, readable outputs, such as displaying the letter 'L' for a left tilt and 'R' for a right tilt, directly on the 7-segment display.

6. Milestones

- Our first objective is to establish the I2C communication protocol, which will enable us to extract raw data from the movement or tilting of the DE10 board
- Next, we will work on developing the Control Unit as well as the Datapath using VHDL.
- Our final milestone is to integrate the design onto the physical DE10-Standard board and verify that the recognized gesture outputs are correctly routed to the 7-segment displays

7. Success Criteria

The project will be a success when the DE10-Standard board can reliably read the accelerometer data, correctly calculate if the board is tilted past our thresholds, and display the correct letters on the 7-segment screen without making mistakes.